Building the SSc-MIM

Construction approach, datasets, and the interaction-validation process

Technical companion to the SSc-MIM overview

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Detailed reference: docs/SSc_MIM_construction_and_validation.md

Three governing principles

1. Deterministic curation

Explicit rules (Mazein 2023 + Disease Maps conventions) so two curators produce the same map. HGNC names, fixed compartments, fixed reaction granularity.

2. Never written speculatively

No edge enters the curated map without a verbatim source quote that passes automated gates AND human ratification.

3. Topology is curated, not learned

Mechanisms come from the literature; patient omics ground, validate and read out the map — they never define its structure.

Notation, format & naming conventions

Notation & format

CellDesigner v4.4 · SBGN Process Description glyphs.

SBML Level 2 Version 4 + CellDesigner annotations.

MI2CAST-compliant annotation on every reaction.

Identifiers (HGNC-anchored)

```
gene/protein  → HGNC symbol  (TGFB1, SMAD3)
proteoform    → symbol+state (SMAD3@S423)
complex       → A:B:C  (SMAD2:SMAD3:SMAD4)
phenotype     → phenotype_<slug>
```

One reaction per ligand–receptor binding event.

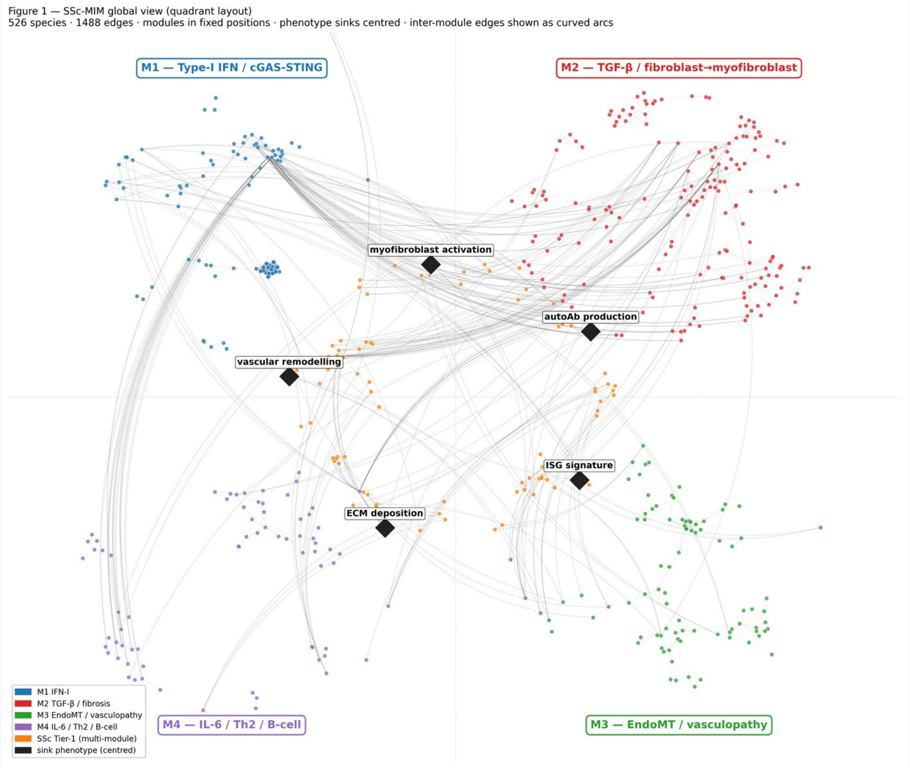
One reaction per kinase–substrate pair (residues collapsed into a state-variable list).

Transcription as TF → mRNA(target), one per target.

Fixed compartment vocabulary (20): extracellular, ECM, plasma membrane, cytosol, nucleus, ER, endosome, mitochondrion, cell-type prefixes…

ARCHITECTURE

Modules, crosstalk & phenotype sinks



M1 Type-I IFN · 19 reactions

M2 TGF-β / fibrosis · 58

M3 EndoMT / vasculopathy · 27

M4 IL-6 / Th2 / B cells · 21

crosstalk · 8

Four phenotype sinks (disease outputs): myofibroblast activation · ECM/collagen deposition · vascular remodelling · ISG signature.

Modelling rule: every entity reaches a sink in ≤ 6 steps — verified, 0 violations (make sink-check).

From literature to a released map

Scope → Reactome import → Harmonise & dedup → Integrate → SSc curation (gated) → Annotate PMID/ECO → QC gates → Release GitHub+Zenodo

Blue = backbone assembly · coral = SSc-specific layer · teal = QC & release.

Each stage is a scripted, reversible step (make targets) with linters and CI — re-running from raw inputs reproduces the map.

The whole automated lane runs end-to-end with `make auto`.

Reactome backbone import & integration

Import canonical pathway scaffolds

M1 type-I IFN · M2 TGF-β (pilot R-HSA-2173789) + PDGF · M3 NOTCH1 (R-HSA-1980143) · M4 IL-6.

Post-process & harmonise

Decode + classify reaction types; rename entities to HGNC; de-duplicate species shared across modules.

Integrate

Merge harmonised module XMLs into one SBML map. Mine embedded PMIDs; auto-fill the bibliography via NCBI E-utils.

```
reactome_pilot.py       (pilot)
post_process_reactome   (harmonise)
harmonise_imports.py    (harmonise)
integrate_modules.py    (integrate)
extract_pmids_*.py      (pmids)
bib_lookup.py           (bib-lookup)
```

Backbone = 159 reactions; gives canonical structure, but propagates curator-inference ECO by default.

STEP 5

The SSc-specific curation layer — the novel content

133 hand-curated reactions Reactome cannot structurally hold (autoantibodies, SSc cell states, GWAS-function, clinical crosstalk). Sourced from a systematic 77-paper worklist (24 themes) + co-author full-text PDFs.

Provenance	n	Meaning
original-curation	40	hand-curated (human-original)
fulltext-verified	27	AI-extracted, verbatim-grounded
ai-discovery (gated)	48	via G0–G4 pipeline
claude-reclassify	10	cell-state → conceptual bridge
claude-lit-mine	8	mined, abstract-verified

```
M2  58   M3  27   M4  21
M1  19   crosstalk  8
```

Every AI-touched row is tagged, reversible (one TSV row), and carries its grounding quote in notes.

126 / 133 carry a primary PMID.

Provenance & graded evidence

The honest denominator: separate the imported backbone from the genuinely SSc-specific layer.

Layer	Rxns	PMID	Exp. ECO
Reactome backbone	159	99%	0
SSc-specific	133	95%	majority

ECO code	Evidence	n
0000314	direct assay	73
0000033	traceable author stmt	27
0000270	expression pattern	20
0000305	curator inference	9
others (353/315)	interaction / mutant	4

122 confirmed · 6 phenotype_aggregation (definitional sinks) · 4 conceptual_bridge · 1 to_complete (POSTN, under expert review).

make evidence-lint fails CI if any SSc reaction is left untriaged — no undeclared inference debt.

Adding an interaction: the edge-discovery protocol

Governing rule: the curated map is never written speculatively.

1. **Stage**
   Candidate edge in staging TSV — with a verbatim supporting sentence from the source paper.
2. **Fetch**
   Cache source text (Europe PMC OA full text / abstract).
3. **Gate**
   Run G0–G4 automated gates → validation_report.tsv.
4. **Ratify**
   A human sets decision = promote (default = held).
5. **Promote**
   Only PASS + promote rows enter the map — tagged, reversible.

Candidates never touch the curated TSV until promoted.

Quality over volume — 12 grounded edges beat 40 shaky ones.

Disease-specific only — content Reactome cannot hold.

One quote → one edge.

Every promoted edge is one removable TSV row.

Five automated gates (G0–G4)

Gate	Check	On failure
G0 schema	required fields present; type in controlled vocabulary	REJECT
G1 HGNC	every gene entity is an official HGNC symbol (HGNC REST)	REJECT (alias→FLAG)
G2 grounding	supporting quote is a verbatim substring of the source text	REJECT — keystone
G3 novelty	gene pair not already an SSc reaction; not a forward Reactome pair	REJECT (dup) / FLAG
G4 evidence	numeric PMID present; corpus text available (SSc context)	FLAG

If the quote is not literally in the paper, the edge is rejected — the core defence against fabricated biology.

Self-test: a fabricated negative-control edge (cand_negctrl, FOXP3→COL1A1, invented quote + decoy PMID) must REJECT on every run; it is hidden from the reviewer app.

Layered QC beyond the gates

Structural

SBML L2V4 validation in CI · sink-connectivity (≤6 steps) · CellDesigner-loadability · MINERVA preflight.

Contradiction detector

Flags any gene pair given opposite polarities by different sources → human arbitration. 0 unresolved.

AI verdict pass

Read each interaction's deciding quote + support/contrary abstracts → validate / revise / caution. Now 128 validate · 5 caution.

Citation-integrity sweep

Caught 28 reactions citing an unrelated PMID; every replacement re-verified vs live PubMed. No PMID cited from memory.

Reviewer app

Offline swipe-deck over 143 interactions: verbatim quote, dossier, module map, advisory verdict; decisions export to CSV/JSON.

Human ratification

Binding step: the expert/co-author adjudicates the biology. AI proposes & self-gates; the human decides.

Four public single-cell datasets

197 donors (121 SSc / 76 HC) — skin, lung, blood — re-analysed from raw GEO files. Citations verified against PubMed.

GEO	Reference (verified)	Tissue	Donors (SSc/HC)
GSE195452	Gur C et al. Cell 2022;185:1373	Skin (multiome)	154 (97 / 57)
GSE138669	Tabib T et al. Nat Commun 2021;12:4384	Skin	22 (12 / 10)
GSE128169	Morse C et al. Eur Respir J 2019;54:1802441	Lung (ILD)	13 (8 / 5)
GSE210395	GEO — no linked publication	PBMC	8 (4 / 4)

Skin cohorts (≈100k + 64k annotated cells) give the fibroblast-subset resolution that M2/M3 model.

Lung (SSc-ILD) + PBMC extend the map beyond skin. Raw archives SHA-256-pinned in data/MIRROR.md (Zenodo mirror).

How the datasets are used

1

Re-process

scanpy QC, Leiden clustering, cell-type annotation from raw counts.

2

Pseudobulk DEG

Per-donor pseudobulk; SSc-vs-HC mixed-effects NB GLM + BH-FDR ($q=0.05$).

3

Map onto species

Match DE genes to map species → coverage.

4

Score per donor

AUCell → each patient = a 4-module activation vector.

5

Overlay

60 MINERVA overlays colour the map by cluster/dataset.

81.3%

of detectable map species seen in patient data

Per-module coverage: M1 84% · M2 88% · M3 75% · M4 71% · Tier-1 83%.

M1 / type-I IFN significantly up in SSc skin (Gur $p=3.2\text{e-}4$; Tabib $p=0.011$).

Cell-type labels validated with CellTypist ($\kappa=0.92$, ARI=0.70). Data grounds the map — it does not define its topology.

Per-donor pseudobulk — the right unit of replication

Tens of thousands of cells, but only 4–13 patients per dataset. The biological question is about patients, not cells.

The problem: pseudoreplication

Cells from one patient are not independent (same genotype, biopsy, batch).

Testing per-cell treats them as thousands of samples → inflated power, many false positives (reviewer R2-M1).

The fix

Rebuild one bulk-like profile per patient → the unit of replication becomes the patient, the correct level for SSc-vs-HC.

```
group cells by (donor × cell_type)
row[gene] = Σ raw counts over cells
```

→ one pseudobulk count row per (donor, cell type).

Then DEG on the pseudobulks

```
NB GLM (pydeseq2) ~ condition
+ Benjamini-Hochberg FDR (q=0.05)
```

Raw counts (not log) — the NB GLM models depth & overdispersion itself.

Trade-off: fewer, more robust hits. Less cell-level granularity, but a valid, reproducible inference — and the same pseudobulks feed AUCell.

AUCell — label-free module activation scoring

“Are a module’s genes concentrated among this patient’s most highly-expressed genes?” If yes → the module is on.

Why — no double-dipping

v1.0 weighted expression by the SSc-vs-HC sign, then tested SSc-vs-HC → circular (reviewer R2-M2).

AUCell scores each donor against the pre-specified module gene set (from the curated map), blind to the SSc/HC label.

Rank-based → robust

Depends only on the order of genes, not absolute values → insensitive to scale, depth, normalisation.

Rank all genes by expression (rank 1 = highest).

Walk the top T (top ~5%); recovery curve R(r) = # module genes with rank ≤ r.

`AUC = Σ R(r),  normalised to [0,1]`

High AUC → module genes cluster at the top → module active.

Output: a 4-module activation vector per patient (+ a Z-score for triangulation).

Feeds the SSc-vs-HC contrasts (M1 IFN p=3.2e-4 in skin) and the endotype workflow (per-cluster module fingerprints).

Scripted, tested, containerised, citable

Scripted pipeline

make auto runs the whole lane; make lint / validate run every content + SBML linter.

Tested + CI

pytest suite (no data needed); GitHub Actions: validate_sbml, lint, scripts-smoke, figures.

Containerised

Dockerfile + workflow pin the environment for end-to-end reproduction.

Provenance & metadata

RO-Crate manifest, MIRIAM injection for BioModels, CITATION.cff + .zenodo.json.

Input-data mirror

data/MIRROR.md + .sha256 — SHA-256-pinned GEO archives (Zenodo mirror).

Release

GitHub + Zenodo DOI at git tag v1.0; make release pre-flights.

In one sentence

A literature-curated map, gated against fabrication, grounded in real patient data

Backbone from Reactome → harmonised → integrated into one SBML map (568 sp / 308 rx / 20 comp).

133 SSc-specific reactions added only through G0–G4 gates + human ratification — no edge without a verbatim source.

Four single-cell datasets (197 donors) ground the map: 81.3% coverage, type-I IFN validated in SSc skin.

Fully scripted, tested, containerised and citable (Zenodo DOI, BioModels-ready).